📄 PRD: AI-Based Medication Non-Adherence System

🧠 1. Product Overview

Product Name: HealMate

Description:
An AI-powered system that monitors patient medication adherence, predicts missed doses, and alerts patients & caregivers to prevent treatment failure.

🎯 2. Problem Statement

Patients often:

- Miss doses
- Take incorrect medication
- Stop treatment early

This leads to:

- Poor health outcomes
- Increased hospitalizations
- Lack of real-time monitoring for doctors

🎯 3. Objectives

- Track medication intake in real-time
- Predict non-adherence using AI
- Provide smart reminders & alerts
- Enable doctors/caregivers to monitor patients

👥 4. Target Users

- Patients (chronic diseases)
- Doctors / healthcare providers
- Caregivers / family members

🚀 5. Key Features

🔹 5.1 Medication Management

- Add medicines (name, dosage, timing)
- Daily schedule view

🔹 5.2 Smart Reminder System

- Adaptive reminders (based on behavior)
- Notifications (mobile alerts)

🔹 5.3 🤖 AI Prediction Engine (Core Feature)

- Predict probability of missing next dose
- Output:
    - Risk Score (Low / Medium / High)

🔹 5.4 📸 Medicine Verification (WOW Feature)

- Camera-based pill detection
- Prevent wrong medicine intake

🔹 5.5 📊 Adherence Dashboard

For patients:

- Daily/weekly adherence %

For doctors:

- Patient compliance score
- Missed dose timeline
- Risk alerts

🔹 5.6 🚨 Alert System

- Missed dose → Reminder
- Multiple misses → Notify caregiver
- Critical → Escalation alert

🔹 5.7 🧠 Behavior Analysis

- Detect patterns:
    - Frequent missed times
    - Weekend skips
- Personalized suggestions

🔹 5.8 🎮 Gamification

- Streak system
- Health score
- Rewards/badges

🧪 6. Data Requirements

- Medication logs (user input)
- Optional:
  - MIMIC-III dataset
  - MEMS dataset

⚙️ 7. Tech Stack

Frontend:

- React / Flutter

Backend:

- Node.js / Firebase

AI/ML:

- Python (scikit-learn / TensorFlow)

Database:

- MongoDB / Firestore

APIs (optional):

- Twilio (SMS alerts)
- OpenCV (image detection)

🧩 8. System Flow (Simple)

1. User enters medication schedule
2. System tracks intake
3. AI analyzes behavior
4. Predicts risk of missing dose

5. Sends smart reminder
6. Updates dashboard
7. Alerts caregiver if needed

📈 9. Success Metrics

- % increase in adherence
- Reduction in missed doses
- User engagement (daily usage)

🎤 10. Demo Plan (VERY IMPORTANT)

1. Add patient + medicines
2. Show daily schedule
3. Simulate missed dose
4. AI predicts risk ⚠️
5. Trigger alert 🚨
6. Show doctor dashboard 📊

👉 Keep demo under **2–3 minutes**